

Storyboard: Bridging the Gap: From Innovation to Patient Impact

Ethics, Evidence, Regulation and Responsible Medical Device Translation • Dr Vikas Kumar Jha • KIIT University, Bhubaneswar • 35–40 minutes (≈35 min scripted + 3–5 min flexible discussion); scripted total 35:00

Story spine. We follow two composite patients, Ramesh (Parkinson's, freezing of gait, FoGO) and Kamala (knee osteoarthritis, SwaKnee), along a six-stage path: Need → Evidence → People → Permission → Access → Safety. Each stage answers one patient question. Every case claim is tagged Demonstrated, Reported (preliminary), Planned or Not established.

Teaching lenses. Stanford Biodesign (need-led process, Principled Decision-Making, Stanford-India Biodesign) • Harvard (HMS Center for Bioethics, MRCT Center, Petrie-Flom, HBS cases) • Cambridge (Engineering Design Centre Inclusive Design Toolkit, Engineering Better Care, Institute for Biomedical Innovation, Judge CCHLE) → adapted to India (CDSCO MDR-2017, ICMR 2017, MvPI, NPPA, UCMPMD 2024, DPDP Rules 2025).

Evidence key Demonstrated Reported, preliminary Planned Not established

SLIDE 1 • OPENING • 0:00–0:45 • 0.75 MIN

Bridging the Gap: From Innovation to Patient Impact

Visual Dark title; authentic photographs of the FoGO ankle module and SwaKnee controller with applicator

Interaction Opening question to the room

Sources Supplied FoGO project material; SwaKnee product photographs



DISCLOSURE

My interests, stated first



I have professional interests in FoGO and SwaKnee.

FoGO founder, Ahilaya Biomedicals; patent filed

SwaKnee [state role]



Vote before my view



Challenge every claim



Protect identities

Sources: University of Cambridge COI policy; Institute of Medicine (2009); Chau, Risken & Roumenau (2010); Stanford GSB Lane OIT 205

2

SLIDE 2 • OPENING • 0:45–1:45 • 1.0 MIN

My interests, stated first

Visual Disclosure panel naming the FoGO roles (founder and PI, patent filed) and a SwaKnee role line for the presenter to complete; three ground rules

Interaction Sets norms: vote first, challenge claims, protect identities

Sources Stanford GSB case on physician COI (2011); HMS COI policy

OPENING VOTE

Would you let a patient use it?



Working prototype



Promising dataset results



Test licence granted

Show 1, 2 or 3 fingers

1

Routine care

2

Research study

3

Not yet

Teaching case, Test Scenario: Form MD-13, Medical Devices Rules 2017

3

SLIDE 3 • OPENING • 1:45–3:45 • 2.0 MIN

Would you let a patient use it? (opening vote)

Visual Three facts (working prototype, promising dataset results, test licence) vs three answer cards (1 Routine care, 2 Research study, 3 Not yet)

Interaction OPENING VOTE by fingers; record split on a flipchart

Sources MDR-2017 (Form MD-13 test licence ≠ permission to treat). The three facts match FoGO's real status (confirmed at the closing vote)

LENSES

Three traditions, one Indian patient



Stanford Biodesign
Start from the need



Harvard bioethics
Respect every person



Cambridge design
Design for whole systems



India

CDSCO

ICMR

MvPI

NPPA

Sources: Stanford Biodesign; HMS Center for Bioethics; MRCT Center, Petrie-Flom; Cambridge EDC, Engineering Better Care, CCHLE

4

SLIDE 4 • OPENING • 3:45–4:15 • 0.5 MIN

Three teaching traditions, one patient in India

Visual Three lens cards converge on an India banner (CDSCO • ICMR • MvPI • NPPA • out-of-pocket care)

Interaction —

Sources Stanford Biodesign process and Need Statements brief; HMS Center for Bioethics, MRCT Center, Petrie-Flom; Cambridge EDC Inclusive Design Toolkit, Engineering Better Care (Clarkson), CCHLE

Two people we will follow



Ramesh, 71

- Parkinson's disease
- Freezes in doorways
- Lives with his daughter



FoGO



Kamala, 64

- Knee osteoarthritis
- 30 km from the clinic
- Pays out of pocket



SwaKnee

Composite personas for teaching; not real patients. Photographs: FoGO alpha prototype; SwaKnee product (company photograph)

5

SLIDE 5 • OPENING • 4:15–5:15 • 1.0 MIN

Two people we will follow

Visual Persona cards: Ramesh (FoGO) and Kamala (SwaKnee) with device photographs

Interaction —

Sources Composite personas (teaching device)

Every stage answers a patient question



Workshop synthesis; adapted from ICHR (2017) principles, the Stanford Biodesign process and Engineering Better Care

6

SLIDE 6 • OPENING • 5:15–5:45 • 0.5 MIN

Each stage must answer a patient's question

Visual Six-node patient journey (editable): Need, Evidence, People, Permission, Access, Safety

Interaction —

Sources Workshop synthesis from ICMR 2017 principles and the Biodesign process

Wording sets the evidence bar



Sources: Stanford Biodesign, Need Statements (2022); MDR 2017; CDSCO Medical Device Software guidance (21 Jul 2020)

7

SLIDE 7 • 1 NEED AND CLAIM • 5:45–7:15 • 1.5 MIN

Claim wording sets the evidence bar

Visual Stanford need-statement card: 'A way to [address problem] in [population] to [achieve outcome]'; two claims → consequences

Interaction Quick call-out: rewrite one claim

Sources Stanford Biodesign Need Statements brief (2022); CDSCO intended-use and software guidance; Simon, Shachar & Cohen, JAMA 2022

EVIDENCE - FoGO

What did the prototype show?

Seen

Not yet shown

Stop sensed

Real freezes

Stop flagged as freeze

Many users

Chest vibrates

Fewer falls

Staged demo - volunteer - simulated freeze

Source: FoGO project material (Willaya Biomedicals), BIRAC BIG, supplied. Staged demonstration: volunteer: simulated freeze

SLIDE 8 • 2 EVIDENCE • 7:15–8:30 • 1.25 MIN

What did the FoGO prototype actually show?

Visual Embedded 12 s authentic prototype clip; Seen (stop sensed → stop flagged as freeze → chest vibrates) vs Not yet shown

Interaction “What did you see? What did you infer?”

Evidence status Demonstrated: staged prototype function on a deliberate stop. Not yet shown: real freezes (vs a voluntary stop), many users, fewer falls

Sources FoGO project deck (BIRAC BIG), supplied

EVIDENCE - FoGO

FoGO has climbed two rungs, not five

Public datasets: F1 0.83–0.85

Prospective patient study

Staged demo, volunteer

Demonstrated

1

Pre-IDEAL

Pre-IDEAL

IDEAL 1-2a

IDEAL 2b-3

IDEAL 4

Benefit in daily life

Not established

Long-term safety and fairness

Not established

5

4

3

2

1

Source: FoGO project material (supplied; project-reported); IDEAL (Lancet 2009; 2019 updates); IDEAL-D (BMJ 2016)

SLIDE 9 • 2 EVIDENCE • 8:30–9:45 • 1.25 MIN

FoGO has climbed two rungs, not five

Visual Evidence ladder with status chips mapped to IDEAL stages

Interaction Which rung matters most before home use?

Evidence status 1 Demonstrated • 2 Reported, preliminary: F1 0.83–0.85 on 3 public datasets (project-reported) • 3 Planned: AIIMS Bhubaneswar PoC • 4–5 Not established

Sources FoGO BIRAC BIG material (supplied); Daphnet (2010); Silva de Lima (2017); IDEAL (2009) / IDEAL-D (2016); Cambridge IBI (2026)

WOULD YOU PROCEED? 1/4

Home pilot for Ramesh next month?

F1 0.83–0.85 on public data

Ten users, alone at home

Ramesh fell twice; asks to join

A Proceed

B Proceed with conditions

C Not yet

Hypothetical scenario. Source: MDR 2017 DMD 13, MD-22/23; KMR (2017); Declaration of Helsinki (2014) para 34

SLIDE 10 • 2 EVIDENCE • 9:45–12:45 • 3.0 MIN

Would you proceed? A home pilot next month

Visual Decision slide: three facts (public-data scores; ten users alone at home; Ramesh fell twice and asks to join) + A/B/C cards

Interaction DECISION 1 (A Proceed • B With conditions • C Not yet); pairs name conditions; does anyone ask about permissions?

Sources ICMR 2017; MDR-2017 clinical investigation; MRCT post-trial access framework (2025)



SLIDE 11 • 2 EVIDENCE • 12:45–14:00 • 1.25 MIN

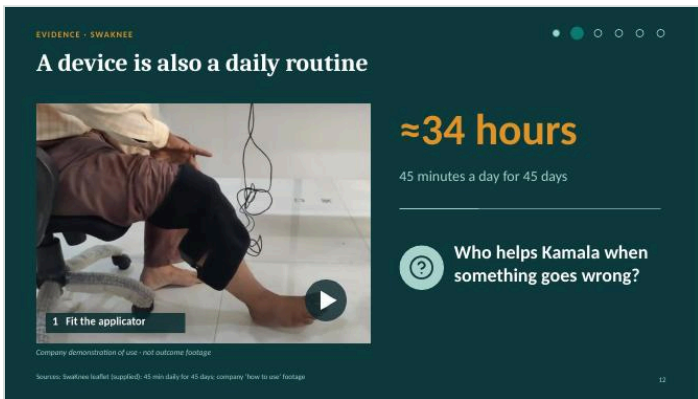
Rank each failure by harm and likelihood

Visual Editable 3×3 risk map with six FoGO failure modes + ‘9–16 false cues per walking hour’ translation of project-reported 0.15–0.27/min (subject-wise to cross-dataset)

Interaction Which control would you test first?

Evidence status Illustrative ranking; no FoGO failure rates implied

Sources ISO 14971:2019; IEC 62366-1; Engineering Better Care; Gerke et al. 2020; Kellmeyer et al. 2016; FoGO project material



SLIDE 12 • 2 EVIDENCE • 14:00–15:15 • 1.25 MIN

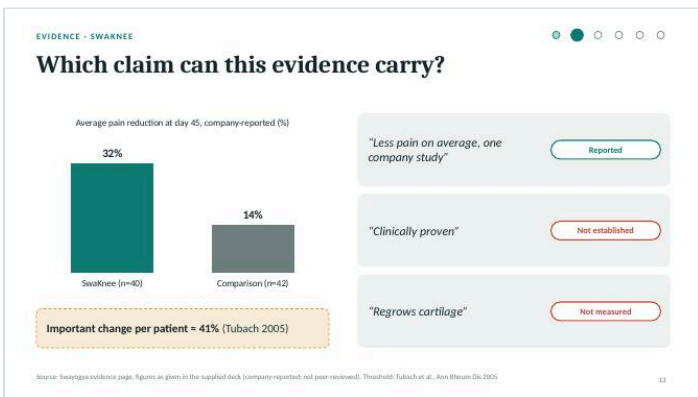
A device is also a daily routine

Visual Embedded 20 s authentic SwaKnee how-to-use clip (fit · switch on · rest); “≈34 hours” stat

Interaction Who helps Kamala when something goes wrong?

Evidence status Company demonstration of use; not clinical-outcome footage

Sources SwaKnee leaflet (45 min/day × 45 days); company video assets



SLIDE 13 • 2 EVIDENCE • 15:15–17:45 • 2.5 MIN

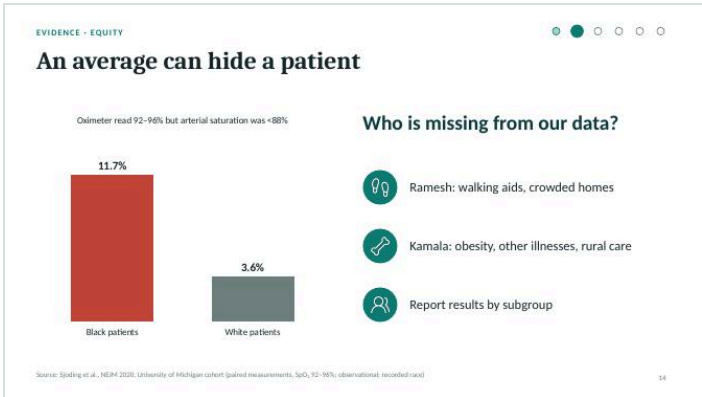
Which claim can this evidence carry?

Visual Native chart 32% vs 14% (company-reported, n=40 vs 42) + patient-important threshold callout (≈41%, Tubach 2005) + claim ladder with status chips

Interaction One defends, one challenges “regrows cartilage”

Evidence status Reported (company, not peer-reviewed) · “Clinically proven” not established · cartilage regrowth not measured

Sources Swayogya evidence page (company-reported); Tubach 2005; Cochrane 2013; Chen 2019; Yang 2020; OARSI 2019; Lundh (Cochrane 2017)



SLIDE 14 • 2 EVIDENCE • 17:45-18:30 • 0.75 MIN

An average can hide a patient

Visual Native chart 11.7% vs 3.6% occult hypoxaemia; “Who is missing?” prompts

Interaction Who is missing from our data? (optional: skip if behind)

Sources Sjoding et al., NEJM 2020; FDA pulse-oximeter draft guidance (Jan 2025); MRCT diversity guidance; Stanford Biodesign JEDI brief

WOULD YOU PROCEED? 2/4

His neurologist is the inventor

“Will saying no change my care?”

Consent form: English only

Only neurologist within 100 km

A Recruit now

B Proceed with safeguards

C Not yet

Hypothetical vignette. Sources: Declaration of Helsinki (2004) para 27; ICMR National Ethical Guidelines (2017); MRCT Center (Harvard) Clinical Research Glossary

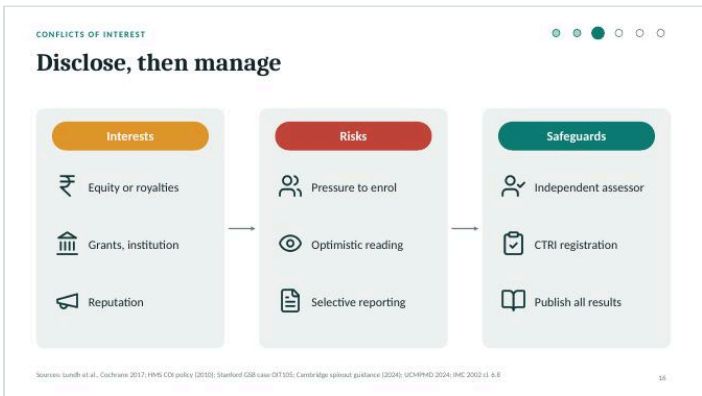
SLIDE 15 • 3 PEOPLE • 18:30-21:30 • 3.0 MIN

Would you proceed? Ramesh’s neurologist is also the inventor

Visual Decision slide with patient quote, English-only consent form, only neurologist within 100 km

Interaction DECISION 2; role-play patient/researcher in 60 s

Sources Declaration of Helsinki 2024 (para 27); ICMR 2017; CIOMS 2016; MRCT Center (Harvard) glossary; US Common Rule key information; Appelbaum (therapeutic misconception)



SLIDE 16 • 3 PEOPLE • 21:30-22:00 • 0.5 MIN

Disclosure starts the work; management finishes it

Visual Interests → Risks → Safeguards infographic

Interaction Which safeguard is cheapest to add today?

Sources Lundh et al., Cochrane 2017; HMS COI policy (2010); Stanford GSB case OIT105; Cambridge spinout guidance; UCMRMD 2024; IMC 2002 cl. 6.8

PERMISSION - INDIA

Each permission answers a different question

Test licence
Make units for testing?
MD-12 → MD-13

Clinical investigation
Test in people?
MD-22 → MD-23 - ethics - CTRI

Manufacturing licence
Sell for this use?
Class A/B: State - C/D: Central

Post-market duties
Still safe?
PSUR - MvPI - recall

FoGO TODAY ✓ MD-13 test licence, 25 units (Aug 2026) → Next: MD-22, ethics, CTRI

Sources: Medical Devices Rules 2017; CDSCO Forms MD-12, MD-22; CDSCO software guidance (2026); FoGO project material (supplied)

17

SLIDE 17 • 4 PERMISSION • 22:00-23:30 • 1.5 MIN

Each permission answers a different question

Visual Four-step CDSCO pathway (test licence → clinical investigation → manufacturing licence, State A/B • Central C/D → post-market) + ‘FoGO today’ ribbon (MD-13 for 25 units, Aug 2026 → next MD-22, ethics, CTRI); US/EU comparison in notes

Interaction Where is Ramesh’s device on this path today?

Sources MDR-2017 (MD-12/13, MD-22/23 Rules 51-52, MD-3/5, MD-7/9); NDCT Rules 2019; CDSCO software guidance (21 Jul 2026); 21 CFR 812; EU MDR; ISO 14155:2026; Kadakia et al. 2024

INCLUSIVE DESIGN

Who can’t use it?

	FoGO	SwaKnee
Vision	Low	Low
Hearing	Low	Low
Thinking	Medium	Medium
Communication	Low	Low
Locomotion	Medium	Medium
Reach and stretch	Medium	Medium
Dexterity	High	Medium

INDIA ADDS

- Language
- Literacy
- Power cuts
- No smartphone
- Caregiver needed

Method: Cambridge Engineering Design Centre, Inclusive Design Toolkit, Ratings Illustrative, India additions, workshop adaptation

18

SLIDE 18 • 5 ACCESS • 23:30-24:15 • 0.75 MIN

Who can’t use it? Audit the demands

Visual Editable inclusive-design audit: seven capability demands (vision → dexterity) rated for FoGO and SwaKnee + ‘Add for India’ panel (language, literacy, power, smartphone, caregiver)

Interaction Pick one demand to redesign (optional: skip if behind)

Evidence status Ratings illustrative, for discussion

Sources Cambridge Engineering Design Centre, Inclusive Design Toolkit; MRCT Accessibility by Design toolkit (2023)

ACCESS

Price is only part of the cost

Buy or rent

Travel

34 hours

Family time

Repairs

Who pays each step?

NPPA PRECEDENT

91%
of home-device brands cut prices

Trade-margin cap, July 2021

Sources: PIB/NPPA trade-margin cap on five home-use devices (13 Jul 2021); Cambridge Judge CCHLE, Journey steps Illustrative

19

SLIDE 19 • 5 ACCESS • 24:15-25:45 • 1.5 MIN

The price tag is only part of the cost

Visual Kamala’s cost journey with five dashed ‘who pays?’ answer slots + NPPA precedent: 91% of home-device brands cut prices after the July 2021 trade-margin cap

Interaction Who pays for each step?

Sources PIB/NPPA trade-margin cap on five home-use devices (13 Jul 2021); knee-implant (16 Aug 2017) and stent (13 Feb 2017) orders in notes; Cambridge Judge CCHLE; HBS Aravind case

WOULD YOU PROCEED? 3/4

Launch with this brochure?

Clinically proven!

Regrows cartilage

Doctor recommended

Avoid surgery forever

HYPOTHETICAL

Distributor insists

Cash lasts 3 months

A Launch as written

B Fix claims first

C Not yet

Hypothetical brochure. Sources: Uniform Code for Marketing Practices in Medical Devices (2024, amended 2026); Drugs and Magic Remedies Act 1954

SLIDE 20 • 5 ACCESS • 25:45–27:45 • 2.0 MIN

Would you proceed? Launch with this brochure

Visual Hypothetical brochure mock-up found by Kamala's son + pressures (distributor insists; cash lasts 3 months) + A/B/C **Interaction** DECISION 3; rewrite one claim to match evidence

Sources UCMPMD 2024 (amended 30 Apr 2026); Drugs and Magic Remedies Act 1954; ASCI Code; OARSI 2019; Stanford Principled Decision-Making

WOULD YOU PROCEED? 4/4

Push tonight's FoGO update?

Fewer false cues

May miss more freezes

Tested on stored data

A Push to everyone

B Staged release with rollback

C Not yet

Hypothetical update. Sources: CDSCO Medical Device Software guidance (2026); US FDA PCDF guidance (Dec 2024)

SLIDE 21 • 6 SAFETY • 27:45–29:15 • 1.5 MIN

Would you proceed? Push tonight's FoGO update

Visual Authentic app screenshot + trade-off facts + A/B/C

Interaction DECISION 4

Sources CDSCO software guidance (2026; Algorithm Change Protocol where applicable); US FDA PCCP guidance (Dec 2024); Gerke et al. 2020; Kellmeyer et al. 2016

SAFETY

When harm happens, who answers?

INDIA

ASR metal-on-metal hips

≈4,700 implanted in India; recalled 2010

1,080 traced by 2018

Report harm: MvPI 1800-180-3024

Sources: MvPI (IPC); ASR; MoHFW expert committee report on reported (2018); PIB compensation formula (29 Nov 2018); Kramer et al. 2013

SLIDE 22 • 6 SAFETY • 29:15–30:30 • 1.25 MIN

When harm happens, someone must answer

Visual Six-node response loop around the patient; ASR card: ≈4,700 implanted in India, 1,080 traced by 2018; MvPI helpline 1800-180-3024

Interaction Assign an owner to each node

Sources MvPI (IPC) and reporting form; MoHFW ASR expert committee (2018); PIB compensation formula (29 Nov 2018); Kramer et al. 2013; Cumberlege 2020

CLOSING VOTE

Would you let a patient use it now?

Which patient?

Which use?

Which evidence?

Who is responsible?

1 Routine care

2 Research study

3 Not yet

Did your vote change? Why?

Compare with the opening vote

23

SLIDE 23 • CLOSE • 30:30–32:00 • 1.5 MIN

Would you let a patient use it now? (closing vote)

Visual Four framing questions + same three answer cards as slide 3

Interaction CLOSING VOTE; compare with opening split; reveal that the case was FoGO (Aug 2026)

Sources —

TAKEAWAY • 90 SECONDS

Commit to one change

Before [next step], I will [action].

Owner: [name]

I proceed only if [condition].



Write it.
Share it.
Keep it.

Adapted from Stanford Biodesign, Principled Decision-Making (2022)

24

SLIDE 24 • CLOSE • 32:00–34:30 • 2.5 MIN

Commit to one change in your own project

Visual Commitment card: Before __ I will __ • Owner • Proceed only if __

Interaction TAKEAWAY EXERCISE (90 s write + 60 s share with neighbour)

Sources Stanford Principled Decision-Making (team values)

REUSABLE TOOL

Patient-impact decision checklist

Stage	Ask before proceeding	Evidence to see	Owner	Go?
Need	Is the need validated with patients and clinicians?	Need statement, observation notes	Clinical lead	☐ Yes ☐ Not yet
Evidence	Does evidence match the claim, users and version? Are failures controlled?	Protocol, subgroups, risk file, version log	Study lead	☐ Yes ☐ Not yet
People	Can people refuse freely? Are conflicts managed?	Consent in local language, COI plan	Investigator, ethics committee	☐ Yes ☐ Not yet
Permission	Which CDSCO permission covers this use?	Class, licence or permission, CTRI	Regulatory lead	☐ Yes ☐ Not yet
Access	Can users afford, use and maintain it? Do claims match evidence?	Total cost, service plan, claims review	Product and service	☐ Yes ☐ Not yet
Safety	Who records, reports and acts on harm?	Complaint log, MvPI route, update plan	Manufacturer, clinic	☐ Yes ☐ Not yet

For every "Not yet": name the owner and the condition for proceeding.

Workshop synthesis — not a regulatory checklist. Copy, adapt and reuse

25

SLIDE 25 • CLOSE • 34:30–35:00 • 0.5 MIN

Patient-impact decision checklist

Visual Editable 6-row table: stage • ask • evidence to see • owner • go/not yet (evidence row covers claim, users, version and risk controls)

Interaction Reusable tool; printed handout

Sources Workshop synthesis

Evidence to request before stronger claims

Question	FoGO	SwaKnee
Outcome	Freeze detection against video-annotated events; falls, confidence, walking	Pain and function with minimal important difference; structural outcomes only if claimed
Comparison	Participant-level validation; cue vs no-cue or sham cue	Randomised, blinded sham comparison; co-interventions recorded
Who	Disease stage, walking aids, homes, turning and doorways	Age, BMI, severity, comorbidities, rural access
Burden and harm	False cues, missed events, signal loss, skin, data	Adverse events, adherence to 45-minute sessions, support needs
Version	Firmware and model version for every result	Controller and applicator version for every result

Prompts for evidence review; not a description of completed work

26

Evidence to request before stronger claims (FoGO vs SwaKnee)

Visual Reference table

Interaction Back-pocket for Q&A

Sources As cited

What independent evidence says

Freezing-of-gait detection and cueing (FoGO)	PEMF for knee osteoarthritis (SwaKnee)
Best-known benchmark: 10 patients, laboratory only (Daphnet, 2010) One review: sensitivity 73–100%, specificity 67–100%, mostly in laboratories Cueing: modest gait gains (RESCUE trial); no trial found showing fewer falls FoGO: F1 0.83–0.85 on public data; prospective study planned (project-reported)	Cochrane 2013 (all OA sites): pain probably eases ~15/100 more than sham Meta-analyses disagree on pain (Chen 2019, knee; Yang 2020, all OA sites) OARSI 2019: electromagnetic therapy strongly recommended against No human evidence found of cartilage regrowth

Sources: Daphnet (2010); Silva de Lima (2017); Newswiser (RESCUE, 2007); Cochrane (2013); Chen (2019); Yang (2020); OARSI (2019)

27

What independent evidence says (FoG detection and cueing; PEMF for knee OA)

Visual Two evidence-context panels

Interaction Back-pocket for Q&A

Sources Daphnet; Silva de Lima; Ginis; RESCUE; Cochrane 2013; Chen 2019; Yang 2020; OARSI 2019; NICE NG226

Teaching materials: Stanford Biodesign and Harvard (1/2)

[Stanford Biodesign, Our Process](#)
Stanford Musculoskeletal Center for Biodesign. Our Process: Identify, Invent, Implement.

[Stanford Biodesign, Need Statements \(2022\)](#)
Stanford Biodesign. Need Statements. Student Guide to Biodesign; 2022.

[Stanford Biodesign, Principled Decision-Making \(2022\)](#)
Stanford Biodesign. Principled Decision-Making. Student Guide to Biodesign; 2022.

[Stanford Biodesign, ITC Brief \(2022\)](#)
Stanford Biodesign. Justice, Equity, Diversity and Inclusion (JEDI). Student Guide to Biodesign; 2022.

[Chao, Bicklin & Krummel \(2010\)](#)
Chao KJ, Bicklin DJ, Krummel TM. A patient-centered, ethical approach to medical device innovation. *Virtual Mentor*. (from AMA J Ethics). 2010;12(2):15–19.

[Stanford GSB case QUTXOS \(2011\)](#)
Burns L, Demondt L, Zenios S. The Role of Physicians in Device Innovation: Critical Success Factor or Conflict of Interest? Stanford GSB case QUTXOS; 2011.

[Stanford Biodesign in India](#)
Stanford Biodesign. Stanford Biodesign in India (Stanford India Biodesign, 2007–2014; now School of International Biodesign, AIIMS and IIT Delhi).

[Chaturvedi et al., BMJ Innovations \(2019\)](#)
Chaturvedi J, Logan A, Narayan G, Kuthappa S. A structured process for unmet clinical need analysis for medical device innovation in India: early experiences. *BMJ Innovations*. 2015;1(2):81.

[Yock, Zenios et al., Biodesign, 2nd ed. \(2015\)](#)
Yock PC, Zenios S, Hahner A, Britten D, Narain SR, Wallens FJ, et al. *Biodesign: The Process of Innovating Medical Technologies*, 2nd ed. Cambridge University Press; 2015 (Stanford Biodesign content).

[Stanford Biodesign, Health Equity](#)
Stanford Biodesign. Health Equity programme and resources.

[HMS Center for Bioethics](#)
Harvard Medical School Center for Bioethics. Fellowship in Bioethics curriculum.

[MRC Center, Diversity guidance \(2020\)](#)
MRC Center of Biogen and Women's Hospital and Harvard. Achieving Diversity, Inclusion, and Equity in Clinical Research: Guidance Document and Toolkit; 2020.

Click a title to open the source. No institution label has evaluated or endorsed FoGO or SwaKnee.

28

Clickable source lists (teaching materials; India; ethics and international; case evidence and media)

Visual Reference slides with hyperlinks

Interaction —

Sources Full list (also in docs/References.md)